IBM Developer
SKILLS NETWORK

# Winning Space Race with Data Science

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14th, Oct 2026



# Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

# Executive Summary

- Objective: Predict whether the Falcon 9 first stage will land successfully and identify factors associated with landing success.
- Methodology: Collected SpaceX launch data using REST APIs and Wikipedia web scraping. Performed data cleaning, exploratory analysis using Python and SQL, interactive visualization with Folium and Plotly Dash, and machine learning classification.
- Key findings: Launch outcomes vary by flight history, launch site, orbit, and payload characteristics. The project evaluates four classification algorithms to identify a suitable landing-success prediction model.
- Business relevance: Predicting booster recovery can support launch planning and estimates of reusable-rocket economics.

# Introduction

SpaceX has reduced launch costs partly through the recovery and reuse of Falcon 9 first-stage boosters. Successful landings are therefore important to operational efficiency.

Research questions:

- Which launch sites have the highest landing success rates?
- How do payload mass, orbit, and flight history affect recovery?
- How has landing performance changed over time?
- Can machine learning predict first-stage landing success?

Section
1

# Methodology



# Methodology

Data collection: SpaceX REST API and Wikipedia scraping.

Data wrangling: Clean missing values, filter Falcon 9 records, and create binary landing labels.

Exploratory analysis: Python visualizations and SQLite queries.

Interactive analytics: Folium maps and Plotly Dash dashboard.

Predictive modeling: Logistic Regression, Support Vector Machine, Decision Tree, and K-Nearest Neighbors.

Evaluation: Hyperparameter tuning, test accuracy, and confusion matrices.

# Data Collection

Two complementary sources were used:

- SpaceX API: Structured launch, rocket, payload, orbit, launchpad, and landing information.
- Wikipedia: Historical Falcon 9 launch records extracted from HTML tables.

Collection pipeline:

Data sources → HTTP requests → JSON/HTML parsing → Pandas DataFrames → Cleaned datasets → Analysis.

# Data Collection – SpaceX API

- Sent GET requests to the SpaceX API.
- Converted JSON responses into Pandas DataFrames using `json_normalize`.
- Extracted rocket, payload, orbit, launch-site, core, and landing information.
- Filtered the resulting dataset to Falcon 9 launches.
- Replaced missing payload-mass values with the mean.
- https://github.com/alim8rby/SpaceX-Falcon9-Data-Science-Capstone/blob/main/notebooks/01_spacex_api_data_collection.ipynb

SpaceX API → GET request → JSON normalization → Extract attributes → Filter Falcon 9 → Clean data → CSV.

# Data Collection - Scraping

- Request the Wikipedia Falcon 9 launch history page.
- Parse HTML using BeautifulSoup.
- Locate launch tables.
- Extract dates, booster versions, launch sites, payloads, orbits, and outcomes.
- Structure records into a DataFrame.
- Clean and export the dataset.
- https://github.com/alim8rby/SpaceX-Falcon9-Data-Science-Capstone/blob/main/notebooks/02_web_scraping.ipynb

SpaceX API → GET request → JSON normalization → Extract attributes → Filter Falcon 9 → Clean data → CSV.

# Data Wrangling

Filtered the dataset to Falcon 9 launches.

Inspected missing values.

Replaced missing payload mass values with the mean.

Created the binary landing-success target variable:

- `1 = successful landing`
- `0 = unsuccessful landing`

Prepared categorical variables for one-hot encoding.

Selected features required for machine learning.

- https://github.com/alim8rby/SpaceX-Falcon9-Data-Science-Capstone/blob/main/notebooks/03_data_wrangling.ipynb

# EDA with Data Visualization

- Scatter plots were used to compare:
  - Flight Number vs. Launch Site
  - Payload Mass vs. Launch Site
  - Flight Number vs. Orbit
  - Payload Mass vs. Orbit
- A bar chart was used to compare landing success rates by orbit.
- A line chart was used to show landing success over time.

**Reason:** These visualizations reveal relationships between mission characteristics and booster landing success.

- https://github.com/alim8rby/SpaceX-Falcon9-Data-Science-Capstone/blob/main/notebooks/04_eda_visualization.ipynb

# EDA with SQL

Queries were used to:

- Retrieve unique launch sites.
- Filter sites beginning with `CCA`.
- Calculate NASA CRS payload mass.
- Calculate average F9 v1.1 payload.
- Find the first successful ground landing.
- Identify successful drone-ship landings between 4,000 and 6,000 kg.
- Count mission outcomes.
- Identify boosters carrying maximum payload.
- Retrieve 2015 drone-ship failures.
- Rank landing outcomes by frequency.

- https://github.com/alim8rby/SpaceX-Falcon9-Data-Science-Capstone/blob/main/notebooks/05_eda_sql.ipynb

# Build an Interactive Map with Folium

Added markers for launch sites.

Used color-coded markers for successful and failed landings.

Used marker clusters to organize overlapping observations.

Added distance lines between launch sites and:

- coastline
- railway
- highway
- nearby city

Used geographic analysis to understand launch-site placement and safety considerations.

- https://github.com/alim8rby/SpaceX-Falcon9-Data-Science-Capstone/blob/main/notebooks/06_folium_launch_sites.ipynb

# Build a Dashboard with Plotly Dash

Dashboard components:

- Launch-site dropdown.
- Payload-mass range slider.
- Pie chart showing landing success/failure.
- Scatter plot showing payload mass vs. landing outcome.
- Color grouping by booster version.

**Purpose:** Allow stakeholders to interactively compare launch-site and payload performance.

- https://github.com/alim8rby/SpaceX-Falcon9-Data-Science-Capstone/blob/main/dashboard/spacex_dash_app.py

# Predictive Analysis (Classification)

**Process**

`Features → One-Hot Encoding → StandardScaler → Train/Test Split → GridSearchCV → Model Comparison → Confusion Matrix`

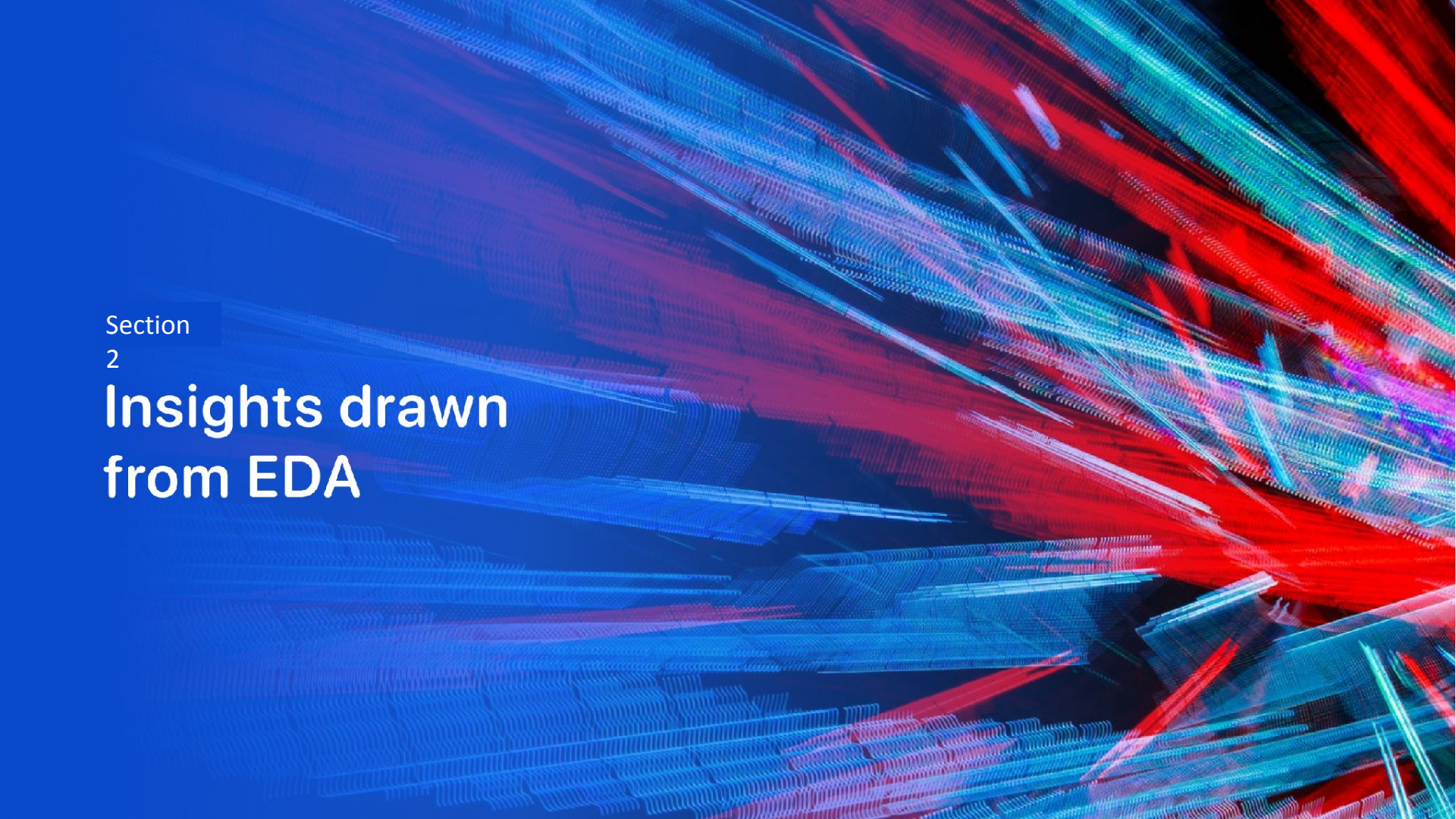
Models:

- Logistic Regression
- Support Vector Machine
- Decision Tree
- K-Nearest Neighbors

- `https://github.com/alim8rby/SpaceX-Falcon9-Data-Science-Capstone/blob/main/notebooks/07_machine_learning_prediction.ipynb`

# Results

- Landing success generally improved as Falcon 9 flight experience increased.
- Success varies by launch site and orbit type.
- KSC LC-39A performed particularly strongly.
- Interactive maps show all major launch facilities are near coastlines.
- Machine learning models achieved approximately **83.33% test accuracy**.

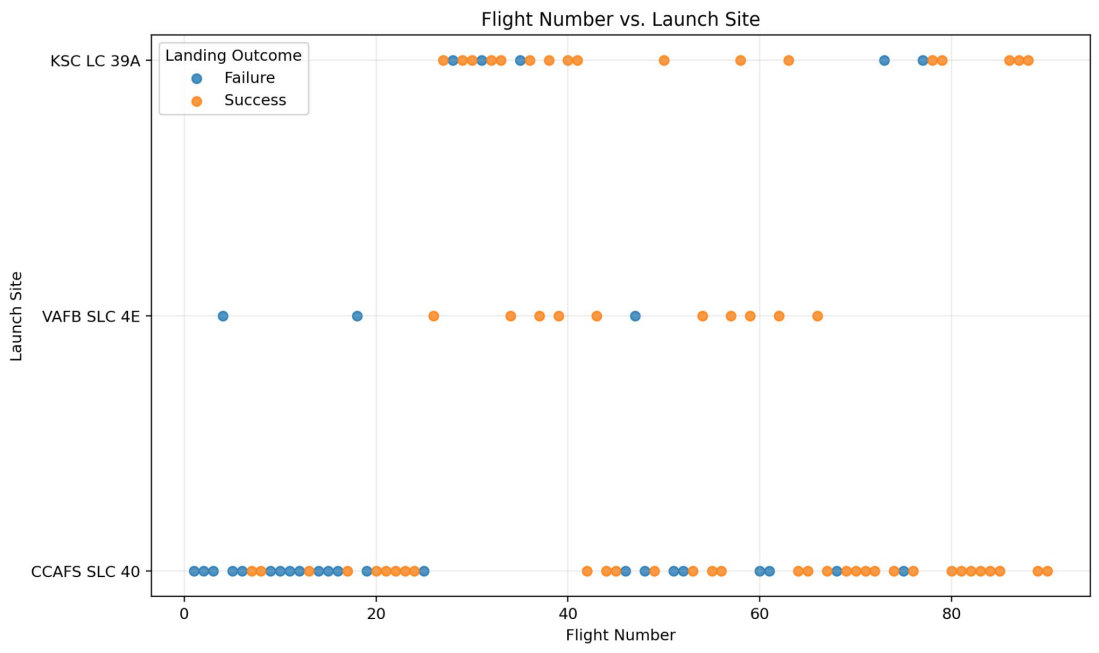


Section 2

# Insights drawn from EDA

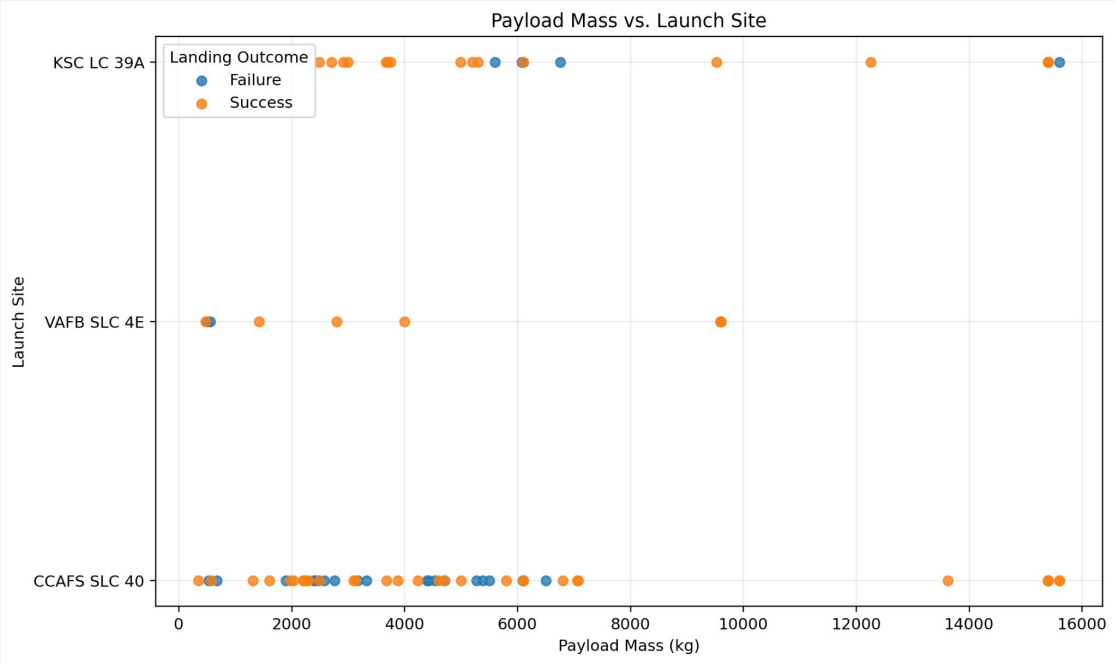
# Flight Number vs. Launch Site

- Landing success tends to become more frequent at higher flight numbers, suggesting that operational experience and improvements in booster technology contributed to better recovery performance.



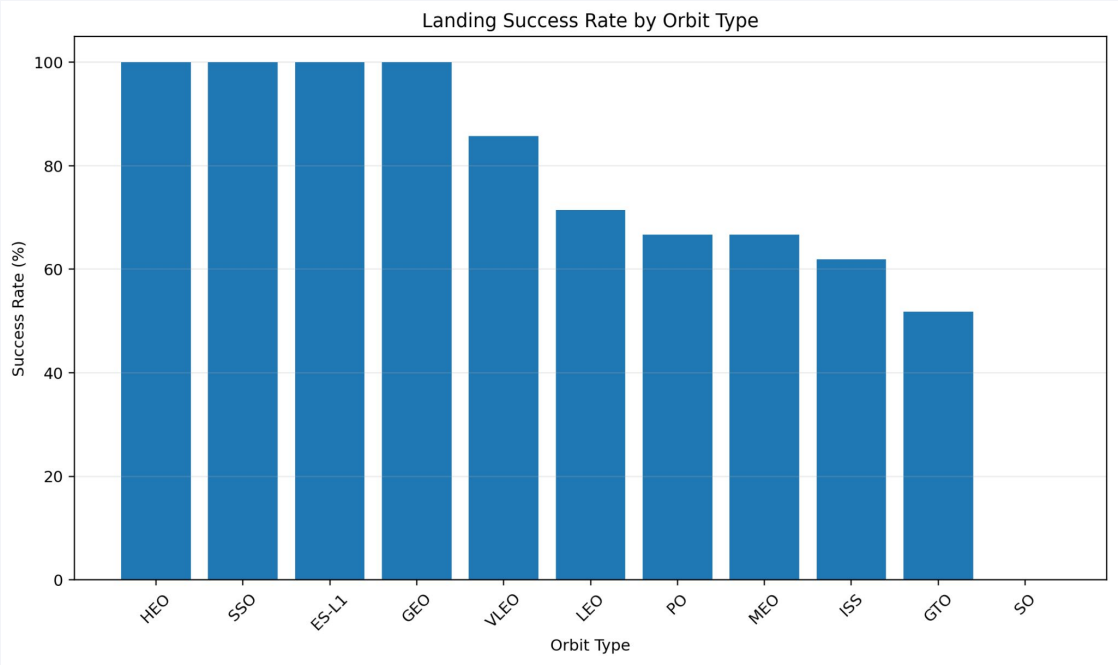
# Payload vs. Launch Site

- Payload mass distributions vary across launch sites. Successful and unsuccessful landings appear across several payload ranges, indicating that payload mass alone does not determine recovery success.



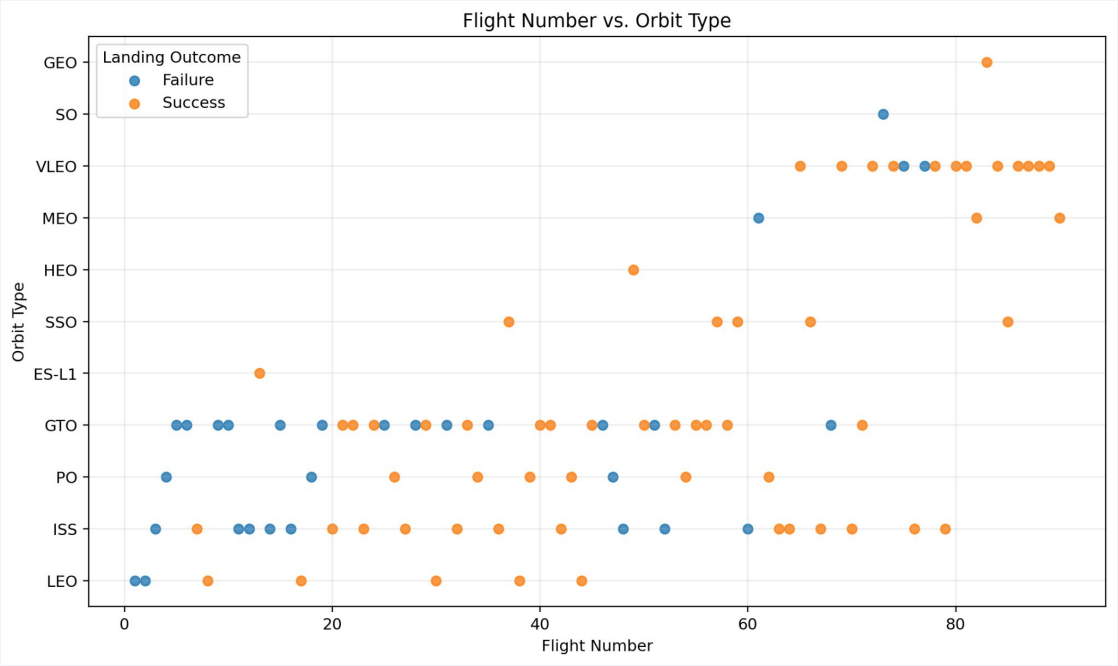
# Success Rate vs. Orbit Type

- Landing success differs substantially by orbit. In common versions of the capstone dataset, **ES-L1, GEO, HEO, and VLEO** show very high or complete success rates.



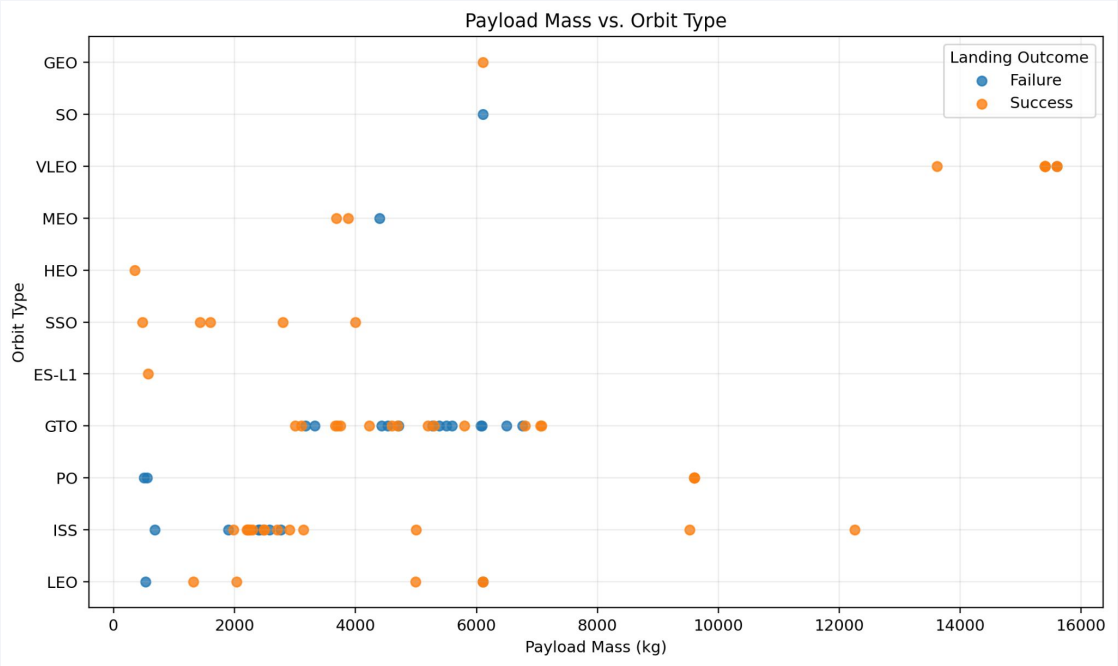
# Flight Number vs. Orbit Type

- Successful landings become more common at later flight numbers for many orbit categories, supporting the idea that SpaceX's recovery capability improved as the program matured.



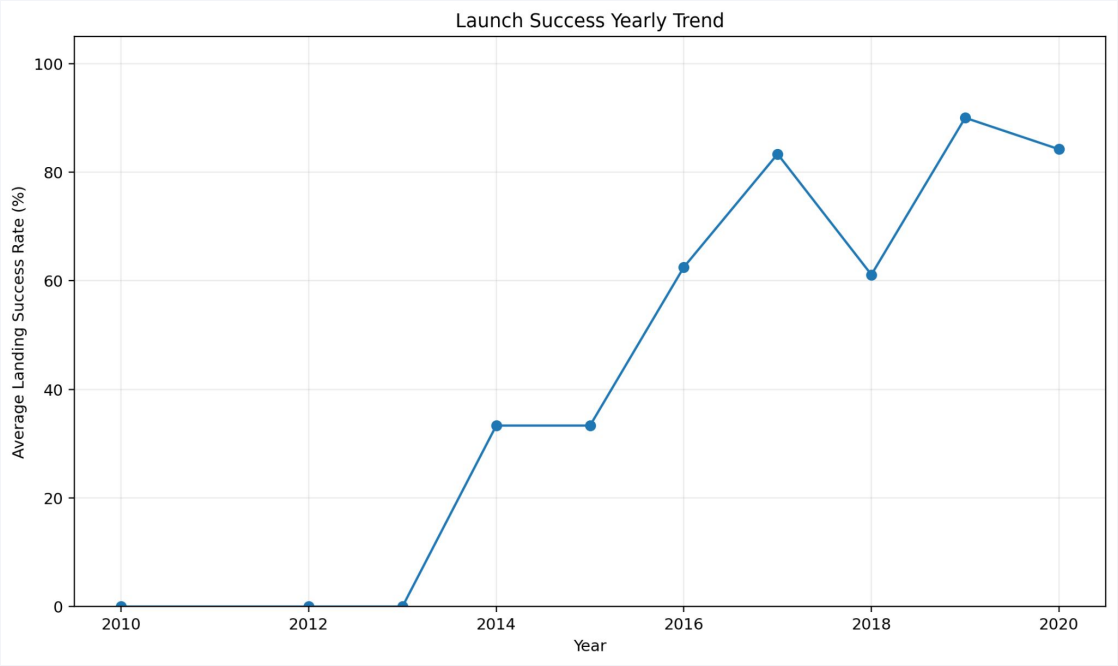
# Payload vs. Orbit Type

- Different orbit types involve different payload profiles. GTO contains both successful and unsuccessful booster recoveries, indicating that mission profile and payload requirements influence recovery difficulty.



# Launch Success Yearly Trend

- Landing success increased markedly from approximately **2013 onward** and reached relatively high levels by the later years of the dataset.



# All Launch Site Names

```
SELECT DISTINCT Launch_Site
FROM SPACEXTBL;
```

**Results**

- CCAFS LC-40
- CCAFS SLC-40
- KSC LC-39A
- VAFB SLC-4E

# Launch Site Names Begin with 'CCA'

```
SELECT *

FROM SPACEXTBL

WHERE Launch_Site LIKE 'CCA%'
LIMIT 5;
```

**Explanation**

This query returns five Falcon 9 records from Cape Canaveral launch sites whose names begin with `CCA`.

# Total Payload Mass

```
SELECT SUM(PAYLOAD_MASS__KG_)
FROM SPACEXTBL
WHERE Customer = 'NASA (CRS)';
```

**Result: 45,596 kg**

**Explanation**

NASA Commercial Resupply Services missions in the dataset carried a combined payload mass of **45,596 kg**.

# Average Payload Mass by F9 v1.1

```
SELECT AVG(PAYLOAD_MASS__KG_)

FROM SPACEXTBL
WHERE Booster_Version LIKE 'F9 v1.1%';
```

A widely used IBM lab version returns approximately **2,534 kg** when including all F9 v1.1 variants.

**Text**

The F9 v1.1 booster family carried an average payload of approximately **2.53 tonnes** in this query definition.

# First Successful Ground Landing Date

```
SELECT MIN(Date)

FROM SPACEXTBL
WHERE Landing_Outcome = 'Success (ground pad)';
```

**Result: December 22, 2015**.

**Explanation**

This marks SpaceX's first successful Falcon 9 first-stage landing on a ground pad.

# Successful Drone Ship Landing with Payload between 4000 and 6000

**Results**

- F9 FT B1022
- F9 FT B1026
- F9 FT B1021.2
- F9 FT B1031.2

**Explanation**

Four booster versions successfully landed on drone ships while carrying payloads between 4,000 and 6,000 kg.

# Total Number of Successful and Failure Mission Outcomes

A common IBM SQLite output gives:

- **Success:** 98
- **Success with whitespace variant:** 1
- **Success (payload status unclear):** 1
- **Failure (in flight):** 1

For a simple slide summary:

**Successful missions: approximately 99–100**
**Failed missions: 1**

The slight difference comes from how string variants are grouped.

# Boosters Carried Maximum Payload

**Maximum recorded payload: 15,600 kg.**

The records correspond to multiple **Falcon 9 Block 5** boosters. Examples include:

- F9 B5 B1048.4
- F9 B5 B1049.4
- F9 B5 B1051.3
- F9 B5 B1056.4
- F9 B5 B1051.4
- F9 B5 B1049.5
- F9 B5 B1060.2
- F9 B5 B1058.3
- F9 B5 B1051.6
- F9 B5 B1060.3
- F9 B5 B1049.7

**Explanation**

Several Block 5 boosters tied for the maximum payload because they carried similarly sized Starlink payload stacks.

# 2015 Launch Records

- Two failed drone-ship landings occurred:

| Date | Booster | Site |
|---|---|---|
| 2015-01-10 | F9 v1.1 B1012 | CCAFS LC-40 |
| 2015-04-14 | F9 v1.1 B1015 | CCAFS LC-40 |

**Explanation**

SpaceX attempted two drone-ship booster recoveries in 2015 in this dataset, and both failed.

# Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

For **2010-06-04 to 2017-03-20**:

1. No attempt — **10**
2. Failure (drone ship) — **5**
3. Success (drone ship) — **5**
4. Controlled (ocean) — **3**
5. Success (ground pad) — **3**
6. Failure (parachute) — **2**
7. Uncontrolled (ocean) — **2**
8. Precluded (drone ship) — **1**

**Explanation**

Early Falcon 9 missions frequently made no recovery attempt, while successful recovery became increasingly common as the program matured.

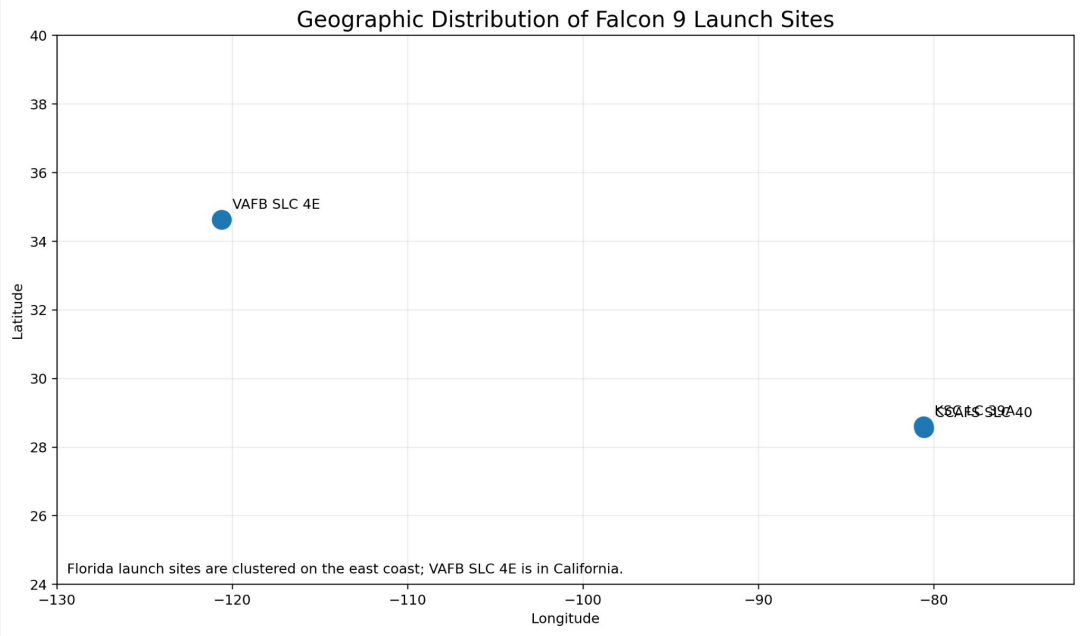
Section 3

# Launch Sites Proximities Analysis



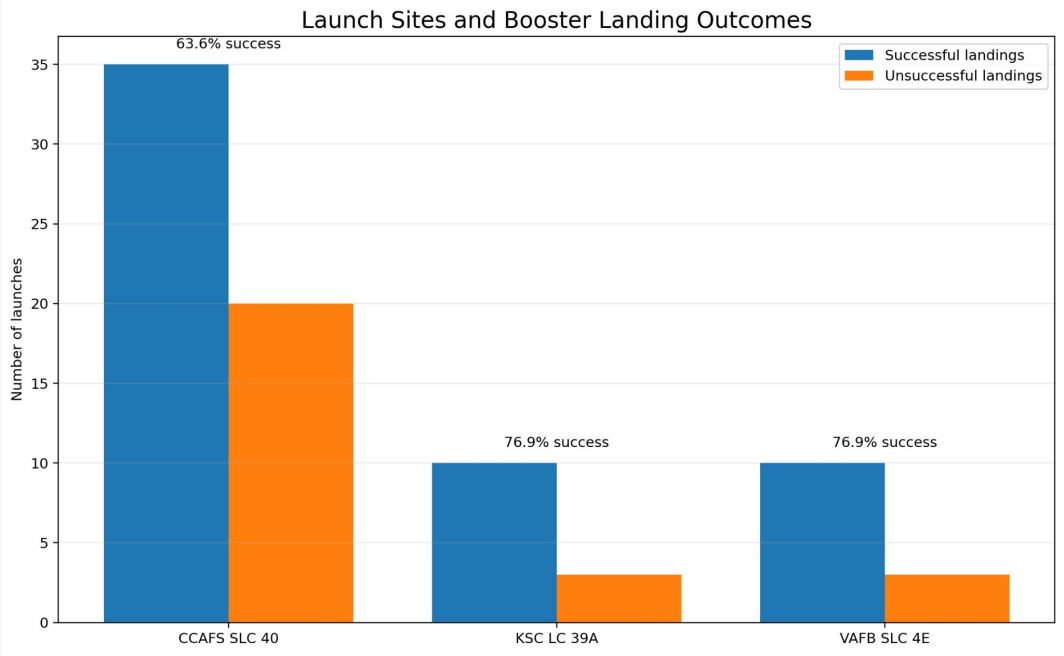
# Geographic Distribution of Falcon 9 Launch Sites

- Three site identifiers are located in Florida and one in California.
- All launch sites are located near the coast.
- Coastal placement supports safe trajectories over open water and reduces risk to populated areas.



# Launch Sites and Booster Landing Outcomes

- Green markers represent successful booster landings.
- Red markers represent unsuccessful landings.
- KSC LC-39A exhibits a relatively high proportion of successful outcomes.

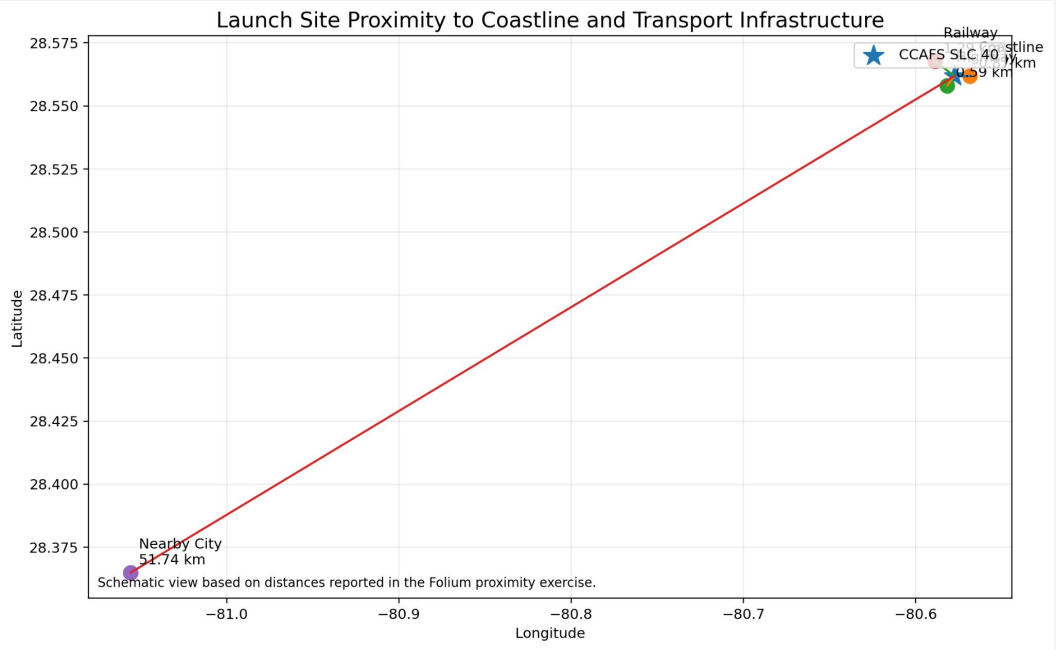


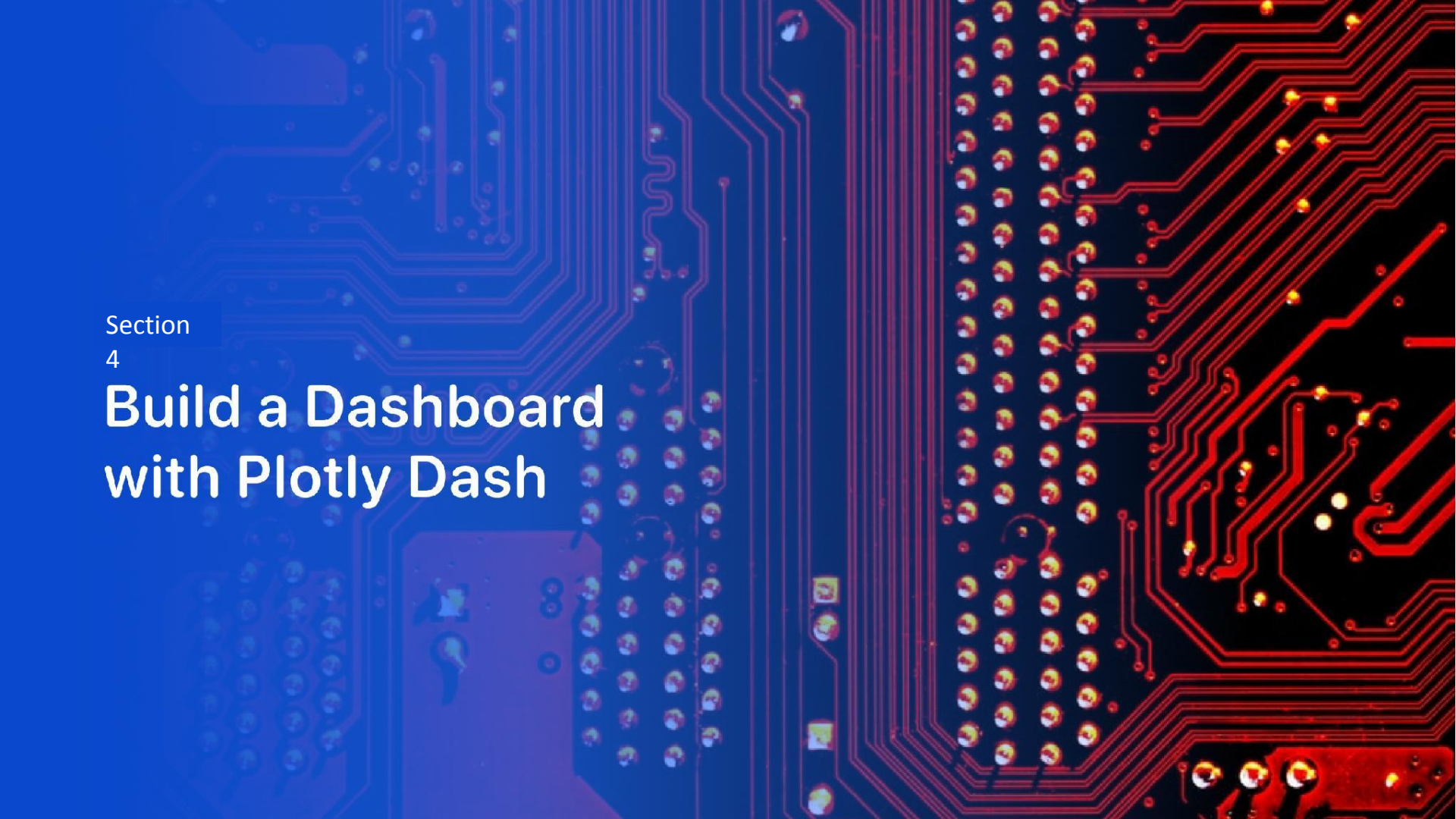
# Launch Site Proximity to Coastline and Transport Infrastructure

- Coastline: approximately **0.87 km**
- Highway: approximately **0.59 km**
- Railway: approximately **1.29 km**
- Closest city: approximately **51.74 km**

**Explanation**

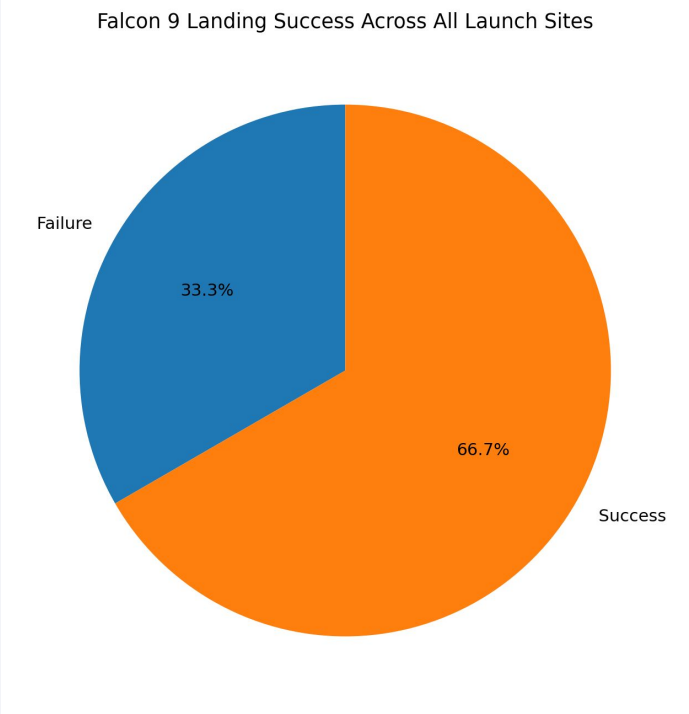
The launch site is extremely close to the coast and transport infrastructure while maintaining a much larger separation from populated urban areas.




Section
4
Build a Dashboard
with Plotly Dash

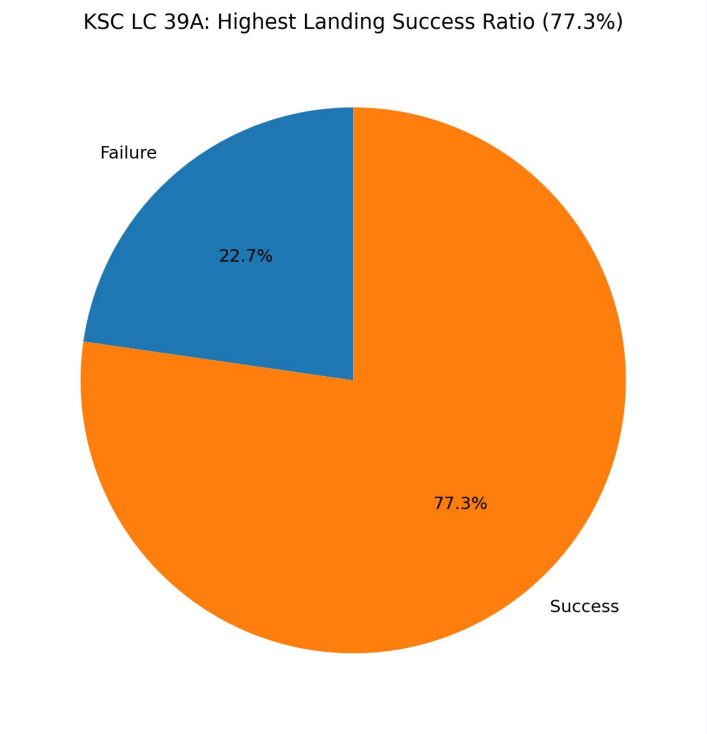
# Falcon 9 Landing Success Across All Launch Sites

KSC LC-39A contributed the largest share of successful launches, accounting for approximately **41.7% of successful launches** in the standard dashboard dataset.



# Launch Site with Highest Landing Success Ratio

KSC LC-39A achieved the highest observed launch success ratio in the dashboard dataset, with approximately 76.9% successful landings.



# Payload Mass vs. Landing Outcome

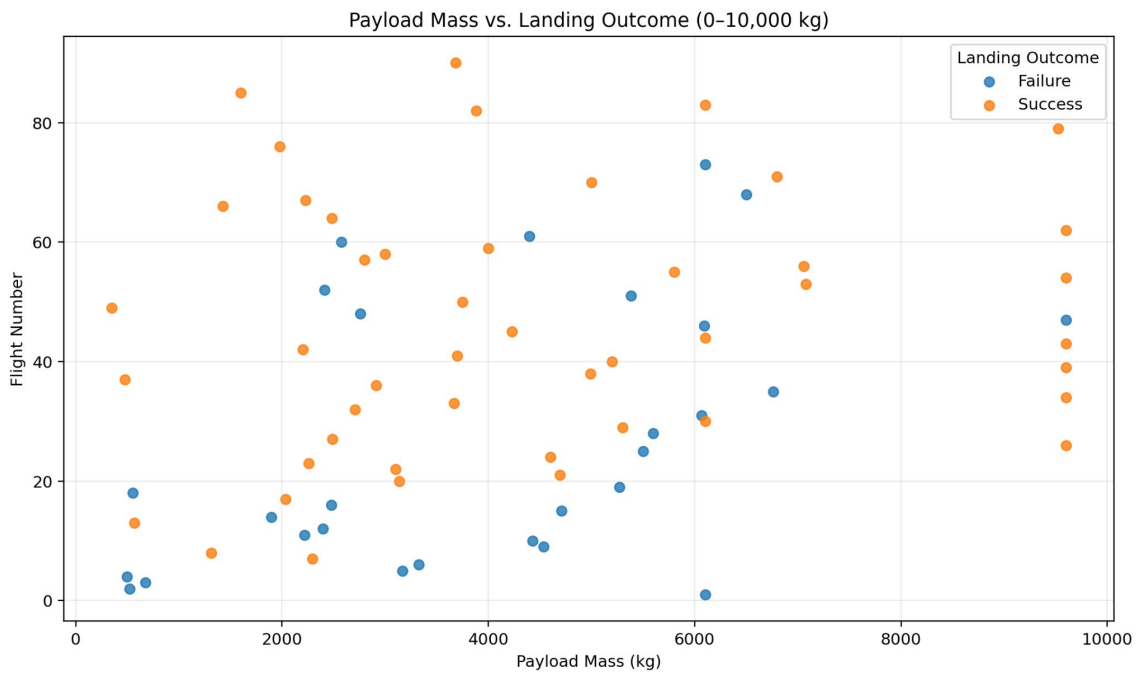
Landing success is not determined by payload mass alone.

A useful comparison is:

- **0–4,000 kg:** lighter payloads
- **4,000–10,000 kg:** heavier payloads

In one standard capstone implementation, the lower payload range showed a higher success proportion than the heavier range.

Booster version also affects performance, indicating that payload and vehicle generation should be analyzed together.



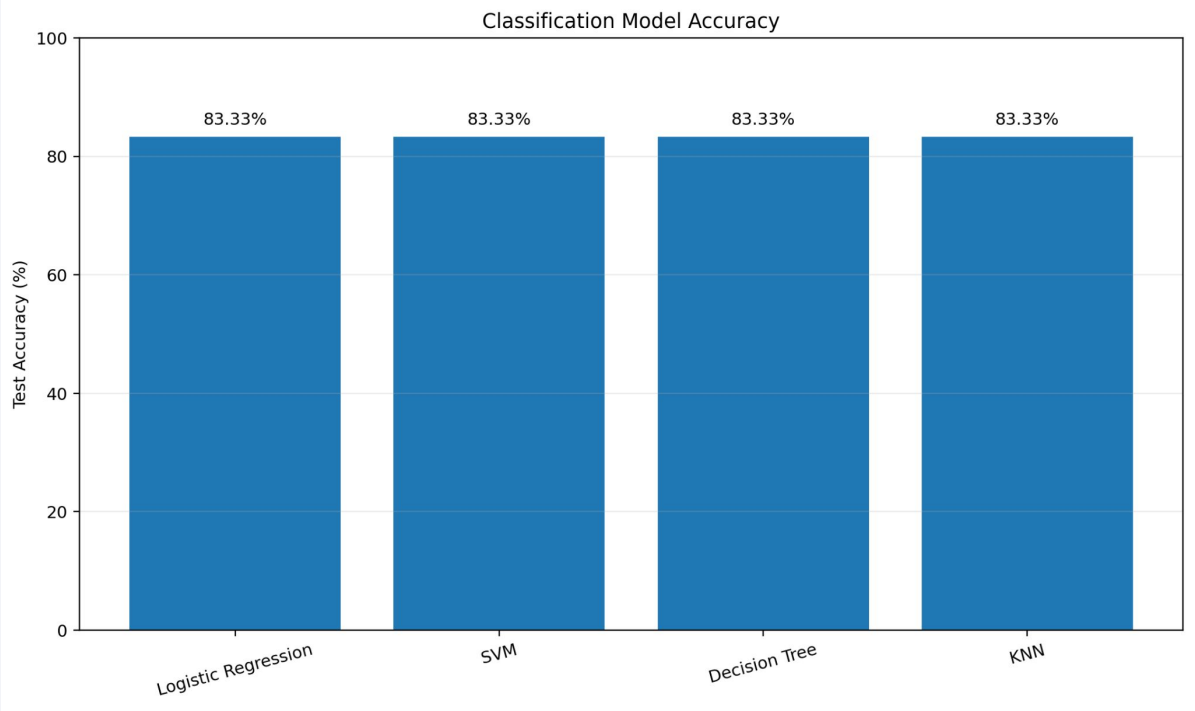
Section
5

# Predictive Analysis (Classification)



# Classification Accuracy

| Model | Test Accuracy |
|---|---|
| Logistic Regression | **83.33%** |
| SVM | **83.33%** |
| Decision Tree | **83.33%** |
| KNN | **83.33%** |



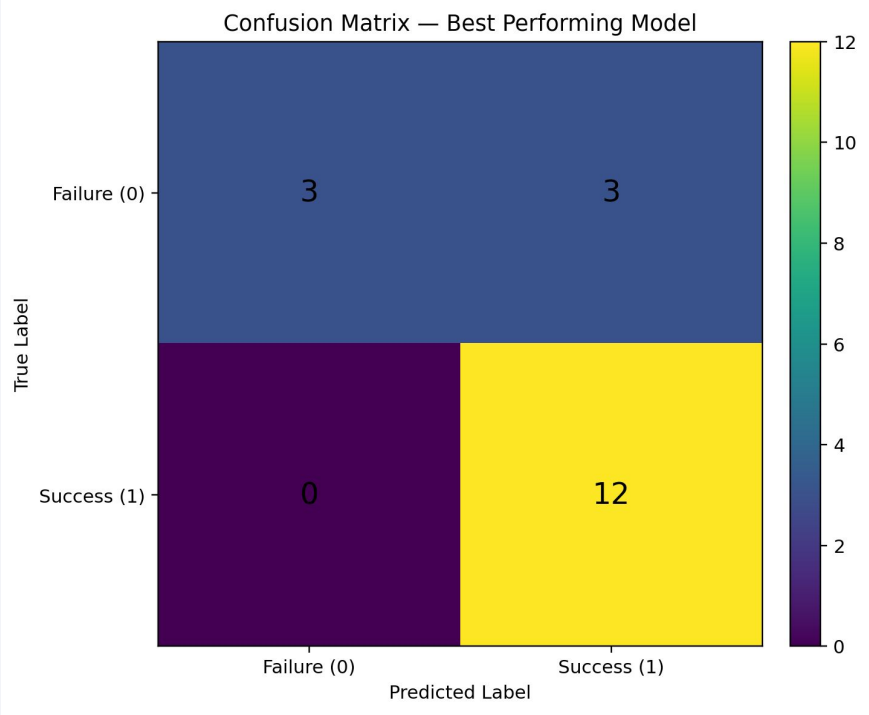

**Explanation**

All four models perform similarly on the small 18-record test set. The Decision Tree can be treated as the strongest overall model when using cross-validation performance as the tie-breaker; published implementations report a GridSearchCV score around **88.9%**.

# Confusion Matrix

Explanation

The model correctly classified 15 of 18 launches. Its main error was false positives: unsuccessful landings predicted as successful.



# Conclusions

- SpaceX's historical Falcon 9 landing performance can be analyzed using publicly available launch data.
- Launch site, orbit, payload mass, and flight history are relevant variables for exploring booster recovery outcomes.
- Visualizations and SQL analysis provide complementary insights into launch performance.
- Folium and Plotly Dash make launch information accessible through interactive maps and dashboards.
- Machine learning provides a framework for predicting first-stage landing outcomes.
- Further evaluation with newer launches and larger datasets is necessary to assess real-world generalization.

# Appendix

**Project Repository**
**`https://github.com/alim8rby/SpaceX-Falcon9-Data-Science-Capstone`**

**Notebooks**
- **SpaceX API Data Collection**
- **Wikipedia Web Scraping**
- **Data Wrangling**
- **EDA with Visualization**
- **EDA with SQL**
- **Folium Launch Site Analysis**
- **Machine Learning Prediction**

**Other Deliverables**
- **Plotly Dash application**
- **SQL queries**
- **EDA visualizations**
- **Folium maps**
- **Model accuracy comparison**
- **Confusion matrix**

# Thank you!

